Department of Indian Scientific Research committee at Bombay (now Mumbai) in Maharashtra. It is the nodal agency for all the research done in the field of atomic energy. Dr. Homi Jahangir Bhaba was the first chairman of Indian Atomic Energy Commission. Now, it is known as Bhaba Atomic Research Centre (BARC).

Nuclear power is the fifth largest source of power in India. Tarapur Atomic Power Station is India's first nuclear power station. Now, there are a total of seven power stations, one each in Maharashtra, Rajasthan, Gujarat, Uttar Pradesh and two in Tamilnadu. In Tamilnadu, we have nuclear power stations in Kalpakkam and Kudankulam. Apsara was the first nuclear reactor built in India and Asia. Now, there are 22 nuclear reactors which are operating in India. Some other operating reactors are

- Cirus
- Dhuruva
- Purnima

Solved problem 6.1

Identify A, B, C, and D from the following nuclear reactions.

(i) $_{13}Al^{27} + A \text{ --------> } _{15}P^{30} + B$

(ii) $_{12}Mg^{24} + B \text{ --------> } _{11}Na^{24} + C$

(iii) $_{92}U^{238} + B \text{ --------> } _{93}Np^{239} + D$

Solution:

(i) $_{13}Al^{27} + _{2}He^{4} \text{ --------> } _{15}P^{30} + _{0}n^{1}$

(ii) $_{12}Mg^{24} + _{0}n^{1} \text{ --------> } _{11}Na^{24} + _{1}H^{1}$

(iii) $_{92}U^{238} + _{0}n^{1} \text{ --------> } _{93}Np^{239} + _{-1}e^{0}$

A is alpha particle, B is neutron, C is proton, and D is electron.

Solved problem 6.2

A radon specimen emits radiation of 3.7×10^{3} GBq per second. Convert this disintegration in terms of curie. (one curie = 3.7×10^{10} disintegration per second)

1 Bq = one disintegration per second

one curie = 3.7×10^{10} Bq

$$1 \text{ Bq} = \frac{1}{3.7 \times 10^{10}} \text{ curie}$$

$$\therefore 3.7 \times 10^{3} \text{ G Bq} = 3.7 \times 10^{3} \times 10^{9} \times \frac{1}{3.7 \times 10^{10}}$$

$$= 100 \text{ curie}$$

Solved problem 6.3

$_{92}U^{235}$ experiences one **α** - decay and one **β** - decay. Find number of neutrons in the final daughter nucleus that is formed.

Solution:

Let X and Y be the resulting nucleus after the emission of the alpha and beta particles respectively.

$$_{92}U^{235} \xrightarrow{\alpha \text{ decay}} {}_{90}X^{231} + {}_{2}He^{4}$$

$$_{90}X^{231} \xrightarrow{\beta \text{ decay}} {}_{91}Y^{231} + {}_{-1}e^{0}$$

Number of neutrons = Mass number – Atomic number

= 231 – 91 = 140

Solved problem 6.4

Calculate the amount of energy released when a radioactive substance undergoes fusion and results in a mass defect of 2 kg.

Solution:

Mass defect in the reaction (m) = 2 kg

Velocity of light (c) = 3×10^{8} m s^{-1}

By Einstein's equation,

Energy released $E = mc^{2}$

So $E = 2 \times (3 \times 10^{8})^{2}$

$= 1.8 \times 10^{17}$ J

Points to Remember

- This phenomenon of spontaneous emission of radiation from certain elements on its own is called 'natural radioactivity'.
- **Curie** is defined as the quantity of a radioactive substance, which undergoes 3.7×10^{10} disintegrations in one second. This is actually close to the activity of 1 g of radium-226.

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